

Forecasting stock return distributions using mixture models

Thomas H. McCurdy

Rotman School of Management, University of Toronto

Administrative Sciences Association 2024 Conference

Abstract

- This presentation will provide examples of theory-informed mixture models which can improve forecasts, providing better guidance for decisions.
- Models (whether cognitive or statistical) are important for separating signals from noise.
- It is important to model higher-order moments of distributions and their interdependence rather than modeling individual moments in isolation.
- I will discuss the power of mixture models for capturing the dynamics (e.g., changing shape) of the distributions and summarize three applications
- The flexible non-linear time-series models that result can be disciplined with parametric structure derived from theory and sensible priors, distinguishing them from black-box nonlinear models.
- Fundamentals, which are treated as latent in the dynamic time-series models, can be used to better understand sources of the dynamics and forecast change points.
- For example, textual analyses converting qualitative information (e.g., news) to quantitative measures (e.g., sentiment) can be useful for capturing the dynamics of the mixture parameters.
- A theoretical representation of behaviour (a nonlinear pricing kernel derived from a generalized preference function) can be used to evaluate to what extent participants care about the dynamics of the stock return distributions (tail risk is priced).

Presentation Outline

- Motivation for models: extracting signals from noise
- Motivation for mixture models

I will summarize three example applications of the mixture distribution approach

- Forecasting the probability of structural breaks
mixture of submodels with different historical data supports
- Forecasting stock returns during the COVID crisis period
mixture of stock market phases: bear market, bear rally, bull market, bull corrections
- Forecasting dynamics in the tail of return distributions
mixture of large (jumps) versus regular (normal) changes in returns
price of jump (skewness) risk is different than price of variance risk

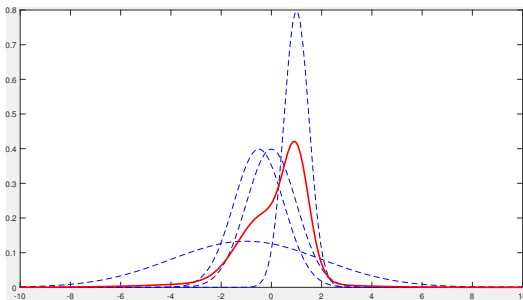
Model Uncertainty and Model Mixtures

Characterizing uncertainty relevant for a particular decision

- Big data – need to separate signal from noise to make good decisions
- In social sciences, parameter and model uncertainty are pervasive
- Result: correlated forecast errors from heterogeneous agents
- Sources of momentum, regime changes, structural breaks, etc.

A finite mixture distribution is a mixture of two or more discrete probability distributions

- Variables are drawn from more than one distribution to create a new distribution
- Example: mixture of normal distributions with different variances and locations



Why Combine Models?

Mix multiple distributions and estimate model weights and transition parameters:

- Rather than assuming that one model will cover all possibilities
- Recognize the limited domain of particular models,
e.g. good versus bad times may involve very different decisions and outcomes
- Choose appropriate models for different ranges of potential outcomes
- Then combine (mix) models with appropriate submodel or regime transition structure
 - Features of the component distributions improve estimates of the aggregate mixture
- Estimate submodel probabilities and model transition probabilities
 - Forecasting transitions (change points) is difficult but fundamentally important
- Update model probabilities as new information arrives (Bayesian learning)
- Powerful method of managing model uncertainty and associated model risk

Traditional Applications of Mixtures

- Markov-Switching (MS) Models are mixtures of discrete states
 - State-specific distributions
 - Estimated transition probability matrix provides the mixing parameters
- Early example of regime-change models for interest rates estimated as MS processes
- Adding intra-regime dynamics
 - Duration-Dependent MS applied to dynamics of GDP, RV, etc.
 - MS-ARCH mixtures applied to volatility regimes, etc.
- GARCH model is a mixture of Normal distributions over time
 - Given $t-1$ information, the conditional density is $N(\mu, \sigma_t^2)$ but the unconditional distribution has fat tails
 - Applied to time-varying quantities (covs, corr, vol) and prices of risk
- Weather forecasting example: mixture of different frequencies

Today's Applications of Mixtures

- *Application 1*: Bayesian averages of submodels which have different data supports
 - Identify and integrate out structural breaks for improved forecasts
 - Identifying optimal data windows for correlation and return distribution forecasts
- *Application 2*: stock market return dynamics
 - Mixture of bear market, bear rally, bull and bull correction states
 - A probabilistic approach to forecasting market dynamics
 - Use the entire distribution to help identify subregimes (bear rallies, bull corrections)
- *Application 3*: Pricing Jump (Tail Risk) Risk
 - The MM GARCH-jump mixture is a mixture of components at t and over time
 - Conditional on knowing # of jumps, the distribution is normal with two GARCH components plus the jump-size component
 - λ_t time-varying jump arrival (function of news) is the mixing parameter
 - Result is a 3-component CPN distribution which varies over time as directed by λ_t
 - Applied with nonlinear pricing kernel to pricing jump (tail) risk, components of risk and return
high frequency news arrival, etc.

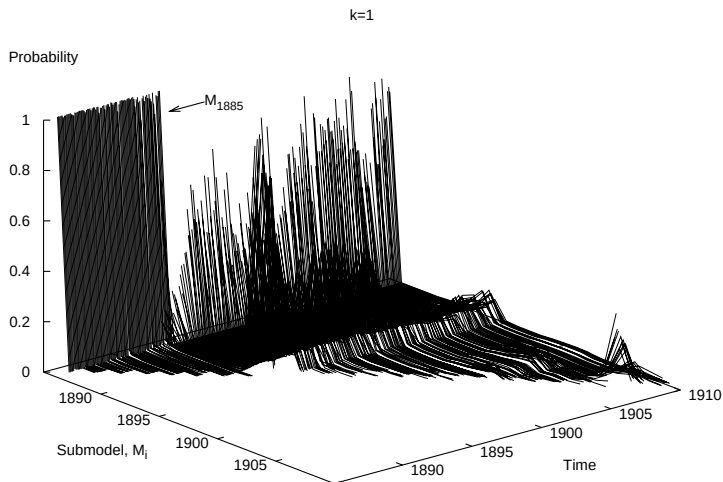
App 1: Using Mixtures to Accommodate Structural Breaks

Focus on potential breaks as source of changes in the unconditional distribution of returns
Forecast distributions using a mixture of submodels – each with a different data support

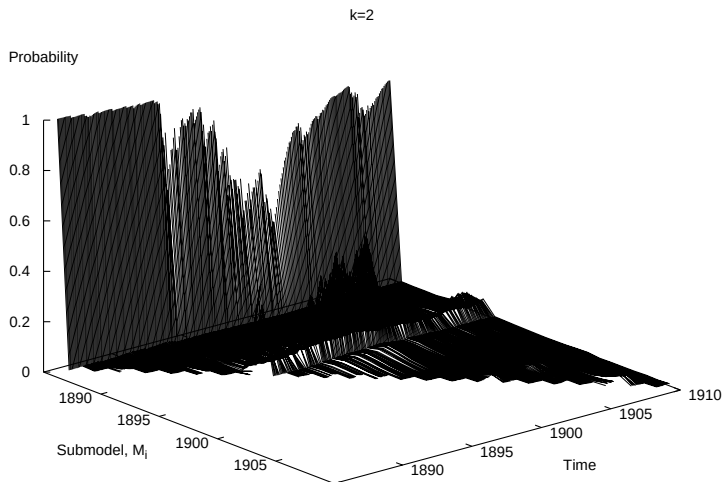
- View the data as being partitioned (by the breaks) into regimes
- New submodels introduced periodically to check for breaks
- Estimate submodel probability weights and the probability of breaks in real time
- Bayesian average of submodels integrates out structural breaks for improved forecasts
- Optimal use of historical data (we show that fixed window is not optimal)
- Probabilistic approach to both past and future breaks
- We learn about breaks over time (time-varying submodel uncertainty)
- Submodel probability distribution provides estimate of most recent break dates
- Real-time distribution forecasts from our break model are superior to alternatives

Reference: 'How useful are historical data for forecasting the long-run equity return distribution?'
(with Maheu), *Journal of Business & Economic Statistics*, January 2009.

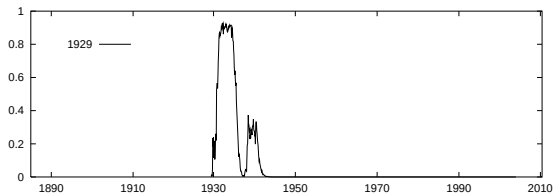
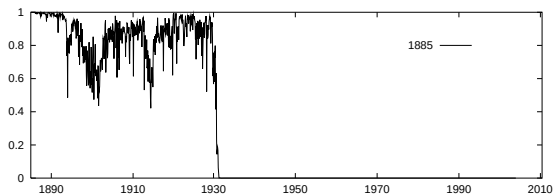
Submodel Probabilities, 1885-1910, ($k = 1$)



Submodel Probabilities, 1885-1910, ($k = 2$)

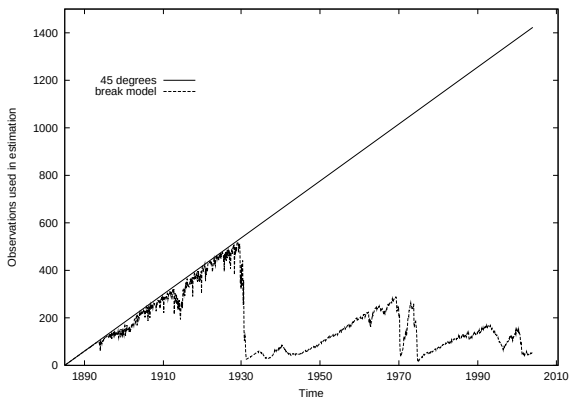


Example: Submodel Probabilities over Time ($k = 2$)



Optimal Data Windows

- The expected number of useful observations for forecasting at each t



- Note that the optimal data window can change at each t
- Also applied this mixture approach to correlation forecasts

Application 2: Forecasting Stock Return Distributions

Applying the mixture model approach to stock market dynamics

- Model the aggregate U.S. stock return dynamics during the recent pandemic period as a mixture of regimes (bear markets, bear rallies, bull markets, bull corrections)
- Demonstrate the model's performance in capturing periods of significant regime change
- Provides forecasts that improve risk management and investment decisions

Our Bull & Bear market model development and applications:

- 'Identifying Bull and Bear Markets in Stock Returns', (with Maheu), *Journal of Business & Economic Statistics* 18 (1), January 2000.
 - Developed a Markov-Switching probability model of the DGP to identify regimes (rather than traditional *ex post* partitioning)
- 'Components of Bull and Bear Markets: Bull Corrections and Bear Rallies' (with Maheu and Song), *Journal of Business & Economic Statistics* 30 (3), July 2012..
 - Added intra-regime dynamics to improve identification of states and improve forecasts for risk management
- 'Bull and bear Markets during the COVID-19 pandemic', (with Maheu and Song), *Finance Research Letters* 42, 2021.

Traditional Bull and Bear Dating Algorithms

Positive and negative trends are typically labelled as 'bull' and 'bear' market regimes

Identified based on an *ex post* assessment of peaks and troughs in the market index

- Pagan-Sossounov [adaptation of Bry-Boschan business cycle algorithm](#)
 - Identify the peaks and troughs by using a window of 8 months.
 - Enforce alternation of phases
 - Eliminate phases less than 4 months unless changes exceed 20
 - Eliminate cycles less than 16 months
- Lunde and Timmermann approach
 - [Use a 6-month window to locate the initial local maximum or minimum.](#)
 - [Bear market begins with a 20% or more decline](#)
 - [Bull market begins with a 20% or more increase](#)

What is Missing?

- *Ex-post* dating schemes fall short in being able guide decisions about regime changes
 - Based on deterministic rules
 - Assume cycles are observable
 - Do not provide probability estimates to guide decisions
 - Cannot make forecasts
- Existing methods of partitioning regimes did not identify intra-regime dynamics that could be useful information for forecasts and decisions
 - Negative sub-trends in a bull market (bull corrections) or
 - Positive sub-trends in bear markets (bear rallies)

Can we identify and forecast stochastic trends and cycles in aggregate equity returns in ways that are useful for investment & risk management decisions?

A 4-State Model for Market Return Dynamics

We propose a nonlinear model to generate probability estimates for market states and distribution forecasts for aggregate stock returns

- We use a mixture distribution: 4 latent states & a sparse transition probability structure
- Include bear rallies and bull corrections to capture heterogeneous intra-regime dynamics
 - Allows transitions from a bear rally either to a bull or back to the bear state
 - Allows transitions from a bull correction to a bear or back to the bull state
- Provides a full probability model of stock market phases
 - The state-specific distributions provide useful information
 - The mixture governs the market dynamics
- Can forecast (market states, risks, and returns) out-of-sample
 - Computation of the expected return takes all 4 possible future states into account
 - State dependent means and other parameters are components of those calculations
 - Conditional VaR predictions are sensitive to market regimes
- Bayesian estimation accounts for parameter and regime uncertainty

Parameterization of the Restricted MS-4 Model

MS-4

$$r_t | s_t \sim N(\mu_{s_t}, \sigma_{s_t}^2)$$

$$p_{ij} = p(s_t = j | s_{t-1} = i), \quad i = 1, \dots, 4, \quad j = 1, \dots, 4.$$

$$\text{Transition matrix } P = \begin{pmatrix} p_{11} & p_{12} & 0 & p_{14} \\ p_{21} & p_{22} & 0 & p_{24} \\ p_{31} & 0 & p_{33} & p_{34} \\ p_{41} & 0 & p_{43} & p_{44} \end{pmatrix}$$

- States refer to s_t and are identified by:

$$\mu_1 < 0 \quad (\text{bear state}),$$

$$\mu_2 > 0 \quad (\text{bear rally state}),$$

$$\mu_3 < 0 \quad (\text{bull correction state}),$$

$$\mu_4 > 0 \quad (\text{bull state});$$

$$\sigma_{s_t}^2 \quad \text{No restriction}$$

- Regimes combine states: $s_t = 1, 2$ bear regime and $s_t = 3, 4$ bull regime
- State transition restrictions
 - Cannot transition from the bear states to a bull correction
 - Cannot transition from the bull states to a bear rally

MS-4-State Model Posterior Estimates

Table: Posterior Estimates

	mean	95% DI
bear μ_1	-0.94	(-1.09, -0.79)
bear rally μ_2	0.23	(0.14, 0.32)
bull correction μ_3	-0.11	(-0.21, -0.02)
bull μ_4	0.52	(0.42, 0.64)
σ_1	5.60	(5.21, 6.03)
σ_2	2.44	(2.27, 2.61)
σ_3	1.85	(1.69, 2.04)
σ_4	1.09	(0.97, 1.21)
μ_1/σ_1	-0.17	(-0.20, -0.14)
μ_2/σ_2	0.10	(0.06, 0.13)
μ_3/σ_3	-0.06	(-0.12, -0.01)
μ_4/σ_4	0.49	(0.35, 0.65)

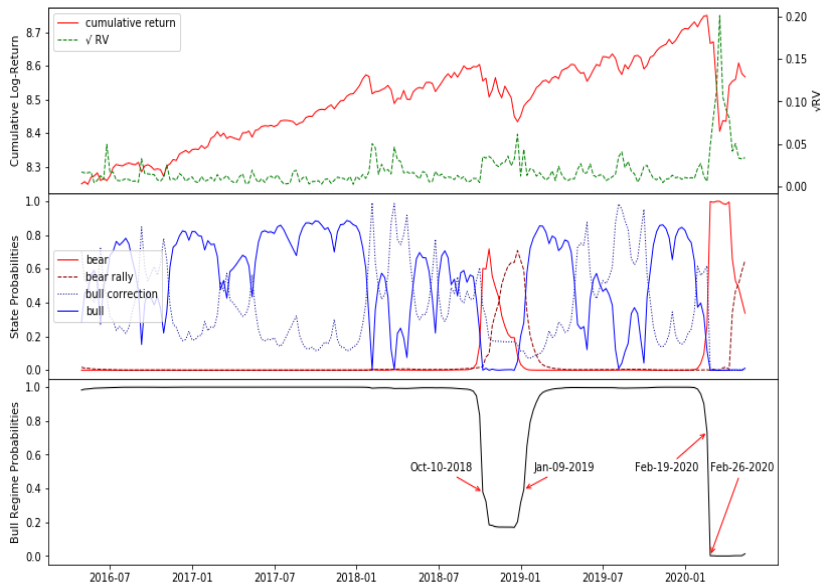
$$\text{Transition matrix } \mathbf{P} = \begin{pmatrix} 0.906 & 0.092 & 0 & 0.002 \\ 0.013 & 0.968 & 0 & 0.019 \\ 0.013 & 0 & 0.891 & 0.097 \\ 0.001 & 0 & 0.122 & 0.876 \end{pmatrix}$$

Unconditional State Probabilities

	mean
bear π_1	0.084
bear rally π_2	0.245
bull correction π_3	0.356
bull π_4	0.316

- Unconditional prob of bear $\pi_1 + \pi_2 = 0.329$
- Unconditional prob of bull $\pi_3 + \pi_4 = 0.671$

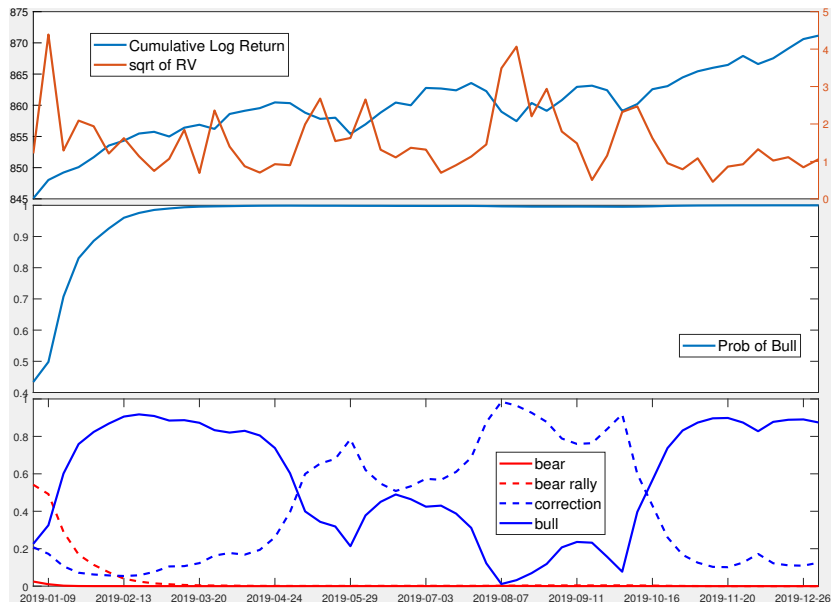
Smoothed Probability Estimates, May 2016-May 2020



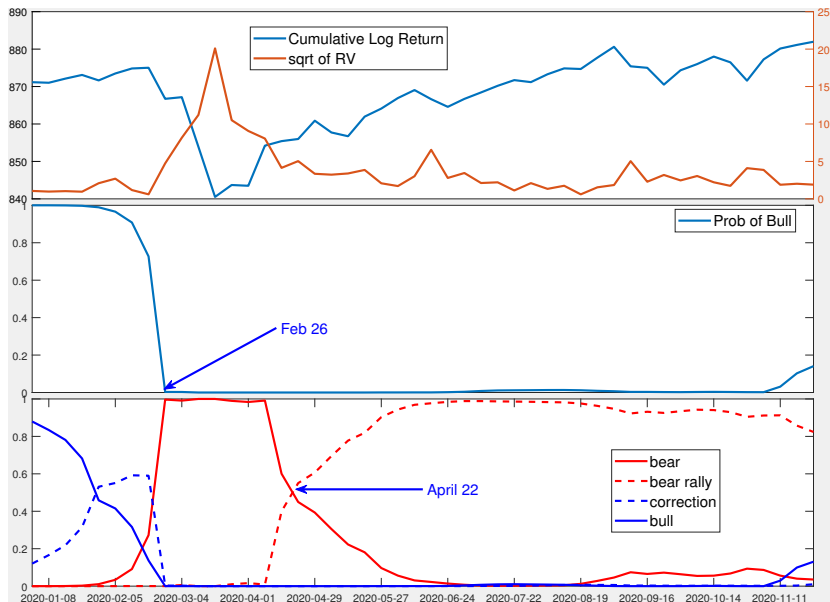
Weekly Stats, Feb-May 2020

week end	wk ret	wk vol	YTD ret	latent state probabilities			
				bear	bear rally	bull correction	bull
1/2/2020	0.55%	1.06%	0.55%	0.000	0.000	0.171	0.829
1/8/2020	-0.15%	0.97%	0.40%	0.000	0.000	0.218	0.781
1/15/2020	1.11%	1.03%	1.51%	0.001	0.000	0.244	0.755
1/22/2020	0.98%	0.96%	2.49%	0.003	0.001	0.352	0.645
1/29/2020	-1.47%	2.09%	1.03%	0.011	0.001	0.616	0.373
2/5/2020	1.86%	2.70%	2.88%	0.040	0.001	0.562	0.398
2/12/2020	1.33%	1.17%	4.22%	0.097	0.002	0.581	0.320
2/19/2020	0.20%	0.61%	4.41%	0.265	0.002	0.618	0.115
2/26/2020	-8.30%	4.74%	-3.89%	0.998	0.000	0.002	0.000
3/4/2020	0.44%	8.16%	-3.45%	0.994	0.005	0.001	0.000
3/11/2020	-13.26%	11.21%	-16.71%	1.000	0.000	0.000	0.000
3/18/2020	-13.38%	20.10%	-30.09%	1.000	0.000	0.000	0.000
3/25/2020	3.18%	10.51%	-26.91%	0.987	0.013	0.000	0.000
4/1/2020	-0.20%	9.07%	-27.11%	0.982	0.018	0.000	0.000
4/8/2020	10.72%	8.05%	-16.40%	0.996	0.005	0.000	0.000
4/15/2020	1.21%	4.14%	-15.19%	0.657	0.342	0.000	0.001
4/22/2020	0.57%	5.04%	-14.62%	0.528	0.470	0.000	0.001
4/29/2020	4.89%	3.35%	-9.73%	0.487	0.512	0.002	0.000
5/6/2020	-3.15%	3.23%	-12.88%	0.413	0.586	0.002	0.000
5/13/2020	-1.00%	3.39%	-13.88%	0.338	0.649	0.002	0.011

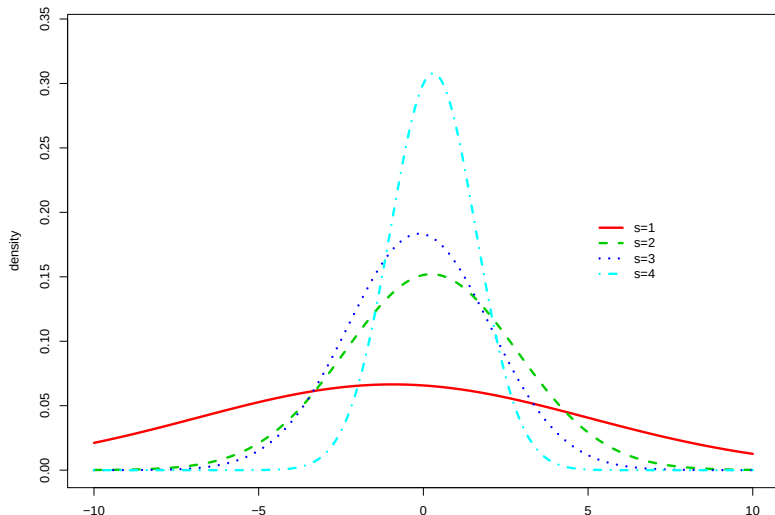
Smoothed Estimates 2019



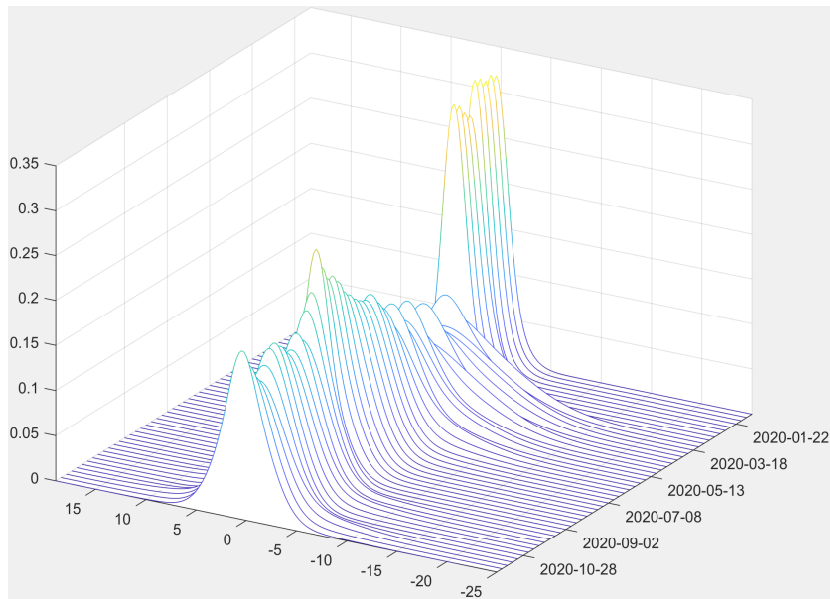
Smoothed Estimates 2020



Distributions for Alternative Market States



Out-of-sample: Predictive Density 1 week ahead



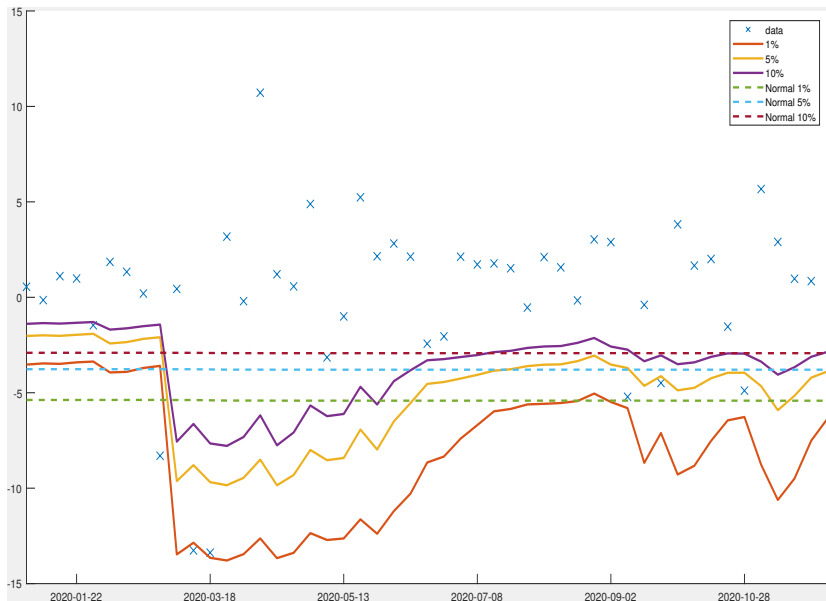
Value-at-Risk (VaR)

- $\text{VaR}_{(\alpha),t}$ is 100 α percent quantile for the distribution of r_t given I_{t-1} .
- Compute $\text{VaR}_{(\alpha),t}$ from the predictive density MS-4 model as

$$p(r_t < \text{VaR}_{(\alpha),t} | I_{t-1}) = \alpha.$$

- Given a correctly specified model, the prob of a return of $\text{VaR}_{(\alpha),t}$ or less is α .
- Comparison with $N(0, s^2)$ where s^2 is the sample variance using I_{t-1} .

Out-of-sample: Value-at-risk 1 week ahead



Investment Returns in 2020

	Return	Sharpe Ratio	number of transactions
Strategy B ^a : $\tau_B = 0.5$	-0.009	-0.048	3
Strategy S ^b : $\tau_S = 0.5$	0.220	1.203	4
Buy-and-hold	0.131	0.566	0

The returns are annualized. Transaction costs are 0.3 cents per share.

^a Buy if $P(B_t = 2 \mid r_{1:t-1}) > \tau_B$ and sell otherwise.

^b Buy if $P(s_t = 2 \mid r_{1:t-1}) > \tau_S$ or $P(s_t = 4 \mid r_{1:t-1}) > \tau_S$. Sell otherwise.

Summary

- Parameterized a 4-state Markov-switching (MS) model for stock returns
 - Offers richer characterizations of market dynamics
 - Two states govern the bear regime
 - Two states govern the bull regime
 - Intuitive restrictions on the state transition probabilities improves forecasts
- Heterogeneous regimes and intra-regime dynamics
 - Allow for bear rallies and bull corrections without a regime change
 - Most regime turning points occur through bear rally or bull correction
 - Volatility dynamics within regimes is important for transitions
 - investors care about both risk and return
- Asymmetric transitions both within regimes and between regimes
 - Bull corrections revert to bull more often than bear rallies bounce back to bear
 - Realized bull and bear regimes can be very different over time
- Probability statements on regimes and future returns available
- Bull corrections and bear rallies empirically important
- Realized bull and bear regimes can be very different over time
- VaR predictions sensitive to market regimes

Motivation Revisited

Can regime probabilities improve risk management & investment decisions?

- Using probabilities to anticipate state or regime transitions could be important for:
 - Investment decisions: rebalance portfolios – e.g. 'take risk off'
 - Risk management decisions: VaR, hedging, etc.

What causes dynamics of the regimes?

- Stay invested and 'weather the storm' for normal fluctuations in returns and risk
- If a bull correction persists and gets deep enough
 - Investors may voluntarily re-allocate portfolios to 'take risk off'
 - Or be forced to do so due to margin calls or factors that kick-in at a threshold
 - These responses could re-inforce the correction & push the market into the bear state with its own reinforcing persistence

Could a nonlinear pricing kernel generate these dynamics?

- A Preference Regime Model of Bull and Bear Markets (Gordon & St-Amour, AER)
- Skewness risk premia based on generalized preferences (Maheu & McCurdy & Zhao, JFE)

Application 3: Pricing Jump (Tail) Risk

Jumps are large price changes:

- Jumps matter for risk management, portfolio selection, pricing, etc.
- We filter returns into large versus smaller changes simultaneously with estimation of all the parameters of the conditional distribution

Are jumps a source of changes in the shape of return distributions?

- Jumps contribute to the conditional variance and drive dynamics in the tail

Do investors care about both the dispersion of their wealth & tail risk for wealth?

- That is, do jumps contribute to the equity premium?

Reference: 'Do jumps contribute to the dynamics of the equity premium?
(with Maheu and Zhao), *Journal of Financial Economics*, 2013.

Dynamics of Higher-Order Moments

In this paper changes in higher-order moments include:

- Dynamics of the conditional variance driven by
 - The direct effect of jumps (jump arrival and jump size)
 - A 2-component asymmetric GARCH process which allows persistence of jumps to be different than normal innovations

Special cases:

- If there are no jumps, conditional skewness is 0 and conditional kurtosis is 3
- If jump arrival is positive but constant, 3rd and 4th moments are non-normal but constant
 - Since 3rd & 4th moments are normalized by a function of conditional variance, conditional skewness and kurtosis will time vary

In our parameterization with time-varying jump arrival:

- All 4 moments of the distribution are time varying

Model of Returns

$$\text{Market Excess Return} = \text{Equity Premium} + \text{Return Innovation} \\ (r = \mathbb{E}[r] + \epsilon)$$

▸ excess return definition

▸ excess return dynamics

Model Part 1/4: Specification for Equity Premium

Our maintained parameterization for the market equity premium is:

- $m_t \equiv$ conditional market equity premium: estimated with & without an intercept μ

$$m_t = \mu + \psi_v v_t + \psi_s s_t + \psi_k k_t$$

- $v_t \equiv$ conditional variance; $s_t \equiv$ conditional skewness; $k_t \equiv$ conditional kurtosis
- Decreasing absolute prudence (Kimball, Econometrica) implies that: $\psi_s \leq 0, \psi_k \geq 0$

In addition to alternative versions discussed in the robustness section, we also estimate

- 1 An unrestricted parameterization of this conditional equity premium; and
- 2 The special case of power utility (Duan and Zhang, 2010)

$$m_t = \mu + \psi_v v_t + \psi_s v_t^{\frac{3}{2}} s_t + \psi_k v_t^2 k_t,$$

in which $\psi_v = (\gamma - 0.5)$, $\psi_s = -(3\gamma^2 - 3\gamma + 1)/6$, $\psi_k = (4\gamma^3 - 6\gamma^2 + 4\gamma - 1)/24$
where γ is defined as the CRRA

Derivation of Test Equation & Parameter Restrictions

Take 4th-order Taylor-series expansion of a general utility function $U(W_{t+1})$, expanding the latter around $W_t(1 + C_t)$, where C_t is an expected return; and derive the pricing kernel approximation:

$$M_{t+1} \approx \sum_{n=0}^3 \frac{U^{(n+1)}(1 + C_t)}{U'(1)n!} (R_{t+1}^W - C_t)^n$$

$$M_{t+1} \approx g_{0t} + g_{1t}(R_{t+1}^W - C_t) + g_{2t}(R_{t+1}^W - C_t)^2 + g_{3t}(R_{t+1}^W - C_t)^3, \quad g_{nt} \equiv \frac{U^{(n+1)}(1 + C_t)}{U'(1)} \frac{1}{n!}, \quad n = 0, 1, 2, 3$$

Preference assumptions (Dittmar 2002): $\text{MU} > 0$, risk aversion, decreasing absolute risk aversion, decreasing absolute prudence, that is, $U' > 0$, $U'' < 0$, $U''' > 0$, $U'''' < 0$ implies that $g_{1t} < 0$, $g_{2t} > 0$, $g_{3t} < 0$

Using $E_t[(1 + R_{t+1}^W)M_{t+1}] = 1$ and $R_t^f \equiv$ the net return on a riskfree asset

$$E_t[R_{t+1}^W] - R_t^f = \kappa_{1t} \text{Cov}_t(R_{t+1}^W, R_{t+1}^W - C_t) + \kappa_{2t} \text{Cov}_t(R_{t+1}^W, (R_{t+1}^W - C_t)^2) + \kappa_{3t} \text{Cov}_t(R_{t+1}^W, (R_{t+1}^W - C_t)^3)$$

in which $\kappa_{nt} = -g_{nt}(1 + R_t^f)$ so that $\kappa_{1t} > 0$, $\kappa_{2t} < 0$, $\kappa_{3t} > 0$, that is,

conditional (co)variance and (co)kurtosis earn positive risk premia and conditional (co)skewness earns a negative premium.

Setting $C_t = E_t[R_{t+1}^W]$, the expected excess return will be a function of the centralized moments, that is,

$$\begin{aligned} E_t[R_{t+1}^W] - R_t^f &= a_t \text{Var}_t(R_{t+1}) + b_t \text{Cov}_t(R_{t+1}, \epsilon_{t+1}^2) + c_t \text{Cov}_t(R_{t+1}, \epsilon_{t+1}^3) \\ &= a_t \text{Var}_t(\epsilon_{t+1}) + b_t E_t[\epsilon_{t+1}^3] + c_t E_t[\epsilon_{t+1}^4] \end{aligned}$$

We standardize the 3rd and 4th conditional moments and relate the estimated coefficients to the preference parameters above

Model Part 2a/4: Diffusive Innovation

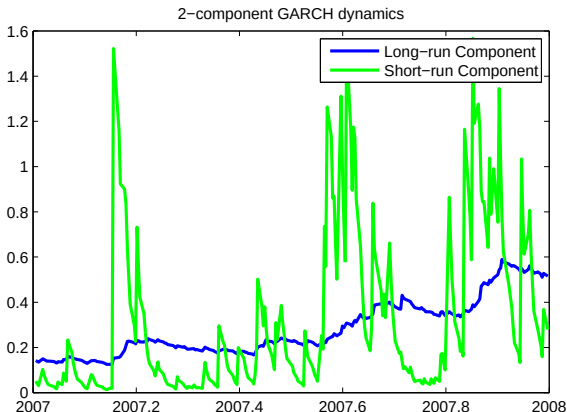
Total return innovation: $\epsilon_{t+1} \equiv \epsilon_{1,t+1} + \epsilon_{2,t+1}$

Diffusive innovation: directed by 2-component GARCH dynamics

$$\epsilon_{1,t+1} | \Phi_t \sim N(0, \sigma_t^2)$$

$$\sigma_t^2 = \text{long-run component} + \text{short-run component}$$

⇒ Captures both **smooth/long-run** component and **transitory/short-run** component



Model 2b/4: The volatility clustering directing process

- The normal innovation to returns, $\epsilon_{1,t+1}$, is assumed to be directed by a 2-component GARCH specification with feedback from jumps

$$\begin{aligned}\epsilon_{1,t+1} &= \sigma_{t+1} z_{t+1}, \quad z_{t+1} \sim NID(0, 1), \quad \epsilon_{1,t+1} | \Phi_t \sim N(0, \sigma_t^2) \\ \sigma_t^2 &= \sigma_{1,t}^2 + \sigma_{2,t}^2 \\ \sigma_{1,t}^2 &= \omega + g_1(A_1, \Phi_t) \epsilon_t^2 + \beta_1 \sigma_{1,t-1}^2 \\ \sigma_{2,t}^2 &= g_2(A_2, \Phi_t) \epsilon_t^2 + \beta_2 \sigma_{2,t-1}^2 \\ \epsilon_t &= \epsilon_{1,t} + \epsilon_{2,t}\end{aligned}$$

- The generalized **news impact coefficient** for the i^{th} GARCH component is

$$\begin{aligned}g_i(A_i, \Phi_t) &= \exp(\alpha_i + I(\epsilon_t)(\alpha_{a,j,i} E[n_t | \Phi_t] + \alpha_{a,i})), i = 1, 2 \\ I(\epsilon_t) &= 1 \text{ if } \epsilon_t < 0, \text{ otherwise } I(\epsilon_t) = 0\end{aligned}$$

- Note that this parameterization incorporates:
 - Asymmetric impact from good versus bad news
 - Asymmetric feedback to future vol from jumps vs normal innovations

Model Part 3a/4: Jump Innovation

Total return innovation: $\epsilon_{t+1} \equiv \epsilon_{1,t+1} + \epsilon_{2,t+1}$

Compensated jump innovation: directed by compound poisson distribution

$n_{t+1} \equiv$ Number of jumps, $Y_{t+1,k} \equiv$ Size of one jump
 $n_{t+1} | \Phi_t \sim \text{Poisson}(\lambda_t), Y_{t+1,k} \sim N(\theta, \delta^2)$

$$P(n_{t+1} = j | \Phi_t) = \frac{\exp(-\lambda_t) \lambda_t^j}{j!}, \quad j = 0, 1, 2, \dots$$

$$\Rightarrow \lambda_t \equiv E[n_{t+1} | \Phi_t]$$

$$\begin{aligned} \epsilon_{2,t+1} &= J_{t+1} - E[J_{t+1} | \Phi_t] \\ &= \sum_{k=1}^{n_{t+1}} Y_{t+1,k} - \theta \lambda_t, \\ \Rightarrow E[\epsilon_{2,t+1} | \Phi_t] &= 0 \end{aligned}$$

Model Part 3b/4: Jump Arrival Dynamics

Jump arrival parameterization:

$$\lambda_t = \gamma_0 + \gamma_1 \lambda_{t-1} + \gamma_2 \zeta_t$$

- The jump arrival *innovation* for period $t - 1$ is defined as

$$\begin{aligned} \zeta_t &\equiv E[n_t | \Phi_t] - \lambda_{t-1} \\ &= \sum_{j=0}^{\infty} j P(n_t = j | \Phi_t, \Theta) - \lambda_{t-1} \\ &= E[n_t | \Phi_t] - E[n_t | \Phi_{t-1}] \end{aligned}$$

- That is, the update in the expected number of jumps for period t when information is updated from $t - 1$ to t
- Jumps are latent: $P(n_t = j | \Phi_t, \Theta)$ is obtained from the estimation filter

Jump arrival has its own dynamics, not necessarily the same as the volatility dynamics

Many models in the literature have specified:

- Time-varying jump arrival (jump intensity) as a function of the diffusive volatility
 For example, $\lambda_t = \gamma_0 \sigma_t^2$
- We will provide some comparisons below

Model Part 4/4: Higher-order Conditional Moments

$$v_t = \sigma_t^2 + \lambda_t(\theta^2 + \delta^2) \equiv \text{conditional variance}$$

$$\text{Recall that } \sigma_t^2 = \sigma_{1,t}^2 + \sigma_{2,t}^2$$

$$s_t = \frac{\lambda_t(\theta^3 + 3\theta\delta^2)}{(\sigma_t^2 + \lambda_t\theta^2 + \lambda_t\delta^2)^{3/2}} \equiv \text{conditional skewness}$$

$$k_t = 3 + \frac{\lambda_t(\theta^4 + 6\theta^2\delta^2 + 3\delta^4)}{(\sigma_t^2 + \lambda_t\theta^2 + \lambda_t\delta^2)^2} \equiv \text{conditional kurtosis}$$

$$m_t = \psi_v v_t + \psi_s s_t + \psi_k k_t$$

$\Rightarrow$ Jump intensity contributes to equity premium both **linearly** and **nonlinearly**

In contrast, for many models in the literature:

- Equity premium as a **linear** function of the jump intensity ($m_t = \psi_v \sigma_t^2 + \psi_j \lambda_t$)
 - Mostly to make the risk neutral pricing tractable

Model Summary

$$r_{t+1} = m_t + \rho_1(r_t - m_{t-1}) + \rho_2(r_{t-1} - m_{t-2}) + \epsilon_{1,t+1} + \epsilon_{2,t+1}$$

$$m_t = \mu + \psi_v v_t + \psi_s s_t + \psi_k k_t, \psi_v \geq 0, \psi_s \leq 0; \psi_k \geq 0$$

$$\epsilon_{2,t+1} = \sum_{k=1}^{n_{t+1}} Y_{t+1,k} - \theta \lambda_t, \quad Y_{t+1,k} \sim N(\theta, \delta^2)$$

$$P(n_{t+1} = j | \Phi_t) = \frac{\exp(-\lambda_t) \lambda_t^j}{j!}, \quad j = 0, 1, 2, \dots$$

$$\lambda_t = E[n_{t+1} | \Phi_t],$$

$$\lambda_t = \gamma_0 + \gamma_1 \lambda_{t-1} + \gamma_2 \zeta_t,$$

$$\epsilon_{1,t+1} = \sigma_{t+1} z_{t+1}, \quad z_{t+1} \sim NID(0, 1)$$

$$\sigma_t^2 = \sigma_{1,t}^2 + \sigma_{2,t}^2,$$

$$\sigma_{1,t}^2 = \omega + g_1(A_1, \Phi_t) \epsilon_t^2 + \beta_1 \sigma_{1,t-1}^2$$

$$\sigma_{2,t}^2 = g_2(A_2, \Phi_t) \epsilon_t^2 + \beta_2 \sigma_{2,t-1}^2$$

$$g_i(A_i, \Phi_t) = \exp(\alpha_i + I(\epsilon_t)(\alpha_{a,j,i} E[n_t | \Phi_t] + \alpha_{a,i})), \text{ for } i = 1, 2;$$

$$I(\epsilon_t) = 1 \text{ if } \epsilon_t < 0, \text{ otherwise } 0$$

Estimation

Likelihood function in closed form
⇒ Estimation done through MLE

► details

Result 1/6: Higher-order Moments in the Equity Premium

$$m_t = \mu + \psi_v v_t + \psi_s s_t + \psi_k k_t$$

Parameter	Pricing Models			
	prudence	variance	skewness	kurtosis
μ		0.020 [3.337]	-0.041 [-2.248]	-0.022 [-1.194]
ψ_v	0.026 [3.396]	0.022 [2.216]	0.045 [4.052]	0.034 [3.151]
ψ_s	-0.027 [-3.423]		-0.069 [-3.015]	
ψ_k	0.000 [0.000]			0.005 [2.295]

- Both variance and skewness are significantly priced
- At the market level skewness and kurtosis pick up the same source of risk
 - Also evident from high negative correlation (-96%) between skewness and kurtosis

► Some more special cases

Result 2/6: Importance of Three Parts in the Model

1. Time varying jump arrival is important
2. 2-component diffusive volatility dynamics is important

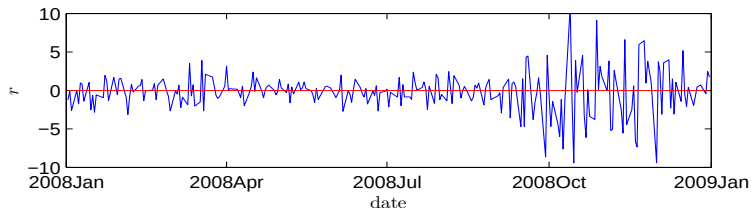
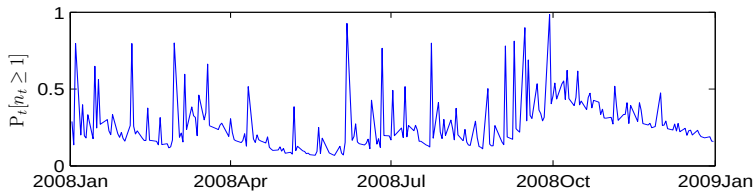
Null hypothesis	LRT Statistic	P-value	Higher moments priced?
Constant jump intensity	83.8	6.34×10^{-19}	No
1-component diffusive volatility	289	2.60×10^{-61}	No

3. Nonlinear pricing of jump intensity is important:

Parameter	Pricing Models	
	our model	linear: $m_t = \psi_v \sigma_t^2 + \psi_j \lambda_t$
ψ_v	0.026 [3.40]	0.017 [1.44]
ψ_s	-0.027 [-3.42]	
ψ_k	0.000 [0.00]	
ψ_j		0.188 [2.47]
lgl	-26386.79	-26392.30

Result 3/6: Diffusive vs Jump Components of the Variance

Sample period	$\bar{\lambda}$	$\bar{v}$	v due to jump part	v due to diffusive part
1926:01–2011:12	0.136	1.054	27%	73%
1926:01–1940:12	0.161	2.130	19%	81%
2008:01–2011:12	0.199	2.813	15%	85%
2008:09–2009:01	0.308	10.167	4%	96%



Result 4/6: Contribution of Jumps to the Equity Premium

Our **time-series** approach with daily **index** returns generates:

- Contribution of jumps to the equity premium about **3.61%** on average
 - Approximately one-third (1.06% per annum) of the jump risk premium comes through contribution of jumps to variance dynamics

$$\psi_v \times (\bar{\lambda} \times (\theta^2 + \delta^2))$$
 - The remainder (2.55% per annum) of the jump risk premium comes through contribution of jumps to skewness
- Skewness premium ($\psi_s \times E[s_t]$) of **3.4%** on average

Application: Out-of-Sample Asset Allocation

Evaluate the out-of-sample portfolio performance with different equity premium models

Utility calibrated to our model and parameter estimates

- Consistent with our assumed pricing kernel

Table: Out-of-Sample Portfolio Performance

Model	Realized Utility		CEQ
	Mean	Std	
Prudence	3.34E-03	5.05E-02	4.005
Variance	3.09E-03	5.73E-02	3.719
GIM-1	2.88E-03	5.51E-02	3.469
Constant	2.80E-03	5.70E-02	3.375

- Realized utility and CEQ improve when pricing higher-order moments (jumps)

Concluding Comments

We show that jumps are priced through higher-order moments in the market return

- More general preferences
- Time-varying jump arrival with own price dynamic
- 2-component diffusive volatility, and
- nonlinear pricing of jump intensity

are **all** important for identifying market prices of **diffusive and jump** risks

Empirically, for our sample, we find:

- Jump risk premia of 3.61% per annum
- Skewness and kurtosis pick up the same tail risk at the market level
 Skewness premium of 3.4% per annum
- Pricing higher-order moments is important for understanding the relationship between equity premium and conditional variance
- Out-of-sample portfolio performance improves when pricing jumps/higher-order moments

Using Textual Analyses of News to Augment Forecasts

Summary of our three applications of mixtures:

- **Application 1:** Bayesian averages of submodels which have different data supports
 - Identify and integrate out structural breaks for improved forecasts
 - Identifying optimal data windows for correlation and return distribution forecasts
- **Application 2:** stock market return dynamics
 - Mixture of bear market, bear rally, bull and bull correction states
 - A probabilistic approach to forecasting market dynamics
 - Use the entire distribution to help identify subregimes (bear rallies, bull corrections)
- **Application 3:** Pricing Jump (Tail Risk) Risk
 - A 3-component CPN distribution which varies over time as directed by jump arrivals
 - Applied with nonlinear pricing kernel to pricing jump (tail) risk
 - Contributes to understanding and forecasting components of risk and return

Extension: replace latent dynamics by textual analyses of high frequency news arrival

Reference: 'News as sources of jumps in stock returns: Evidence from 21 million news articles fro 9000 companies' (with Jeon and Zhao), *Journal of Financial Economics*, 2022.

Discussion: forecasting with nonlinear mixture distribution parameterizations vs machine learning

Appendix: Definition of Excess Returns

Define the continuously compounded excess return on the market index as

$$r_t \equiv r_{m,t} - r_{f,t}$$

$r_{m,t} \equiv$ the continuously compounded market index return

$r_{f,t} \equiv$ the continuously compounded riskfree rate

Henceforth, we refer to r_t as the log return

Appendix: Dynamics of Returns

Assume that the dynamics of continuously compounded market excess returns are driven by

$$r_{t+1} = m_t + u_{t+1}$$

$m_t \equiv$ expected market equity premium using info at time t

$$u_{t+1} \equiv \rho_1 u_t + \rho_2 u_{t-1} + \epsilon_{t+1}$$

$\epsilon_{t+1} \equiv$ mean zero return innovation

That is, excess returns are driven by:

- The expected market equity premium m_t
- An autoregressive component due to stale prices or missing pricing factors
Note: we test for robustness to including or excluding this AR structure
- And a mean zero return innovation ϵ_{t+1}

Appendix: Maximum Likelihood Estimation

Building the likelihood function in closed form:

Given the number of jumps $j \in \{0, 1, 2, \dots\}$ and parameter vector Θ

the conditional density of returns $f(r_t | \Phi_{t-1}, \Theta, n_t = j)$ is Normal:

$$\frac{1}{\sqrt{2\pi(\sigma_t^2 + j\delta^2)}} \exp\left(-\frac{1}{2} \frac{(r_t - m_t - \rho_1(r_{t-1} - m_{t-1}) - \rho_2(r_{t-2} - m_{t-2}) - (j - \lambda_t)\theta)^2}{\sigma_t^2 + j\delta^2}\right)$$

The full likelihood contribution in terms of r_t is then

$$f(r_t | \Phi_{t-1}, \Theta) = \sum_{j=0}^{\infty} f(r_t | \Phi_{t-1}, \Theta, n_t = j) P(n_t = j | \Phi_{t-1}, \Theta)$$

in which $P(n_t = j | \Phi_{t-1}, \Theta)$ is the p.d.f. of the time-varying Poisson distribution

The full-sample loglikelihood $l = \sum_{t=1}^T \log f(r_t | \Phi_{t-1}, \Theta)$ is maximized wrt Θ

The filter is:

$$P(n_t = j | \Phi_t, \Theta) = \frac{f(r_t | n_t = j, \Phi_{t-1}, \Theta) P(n_t = j | \Phi_{t-1}, \Theta)}{f(r_t | \Phi_{t-1}, \Theta)}, j = 0, 1, 2, \dots$$

Appendix: Special Cases for Equity Premium Specification

Parameter	Pricing Models			
	GIM-1	GIM-2	constant λ	no $AR(2)$
μ	0.028	0.019	0.010	-0.053
	[4.241]	[2.938]	[0.806]	[-2.952]
ψ_v	0.001	0.023	0.017	0.043
	[0.123]	[2.690]	[1.652]	[4.213]
ψ_s			-0.013	-0.107
			[-0.981]	[-3.660]
ψ_k			0.000	0.000
			[0.000]	[0.000]